MnemoBench: Evaluating Belief Update, Self-Pollution Resistance, and Forgetting in Long-Lived Memory Systems

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Abstract

Persistent memory layers for LLM agents are typically evaluated on *recall*: can the system retrieve what was said earlier? We argue that in long-lived use the dominant failure modes are not recall but *maintenance*: (1) failing to revise a belief when a fact changes, (2) letting the agent's own generated or imagined content pollute the user's fact base, and (3) never forgetting, so that stale trivia degrades precision over time. We introduce **MnemoBench**, a reproducible benchmark with three task families targeting exactly these behaviours, with ground truth fixed by a generator rather than judged post hoc. We evaluate Nemos, an embeddable memory kernel with bitemporal contradiction invalidation, namespace isolation between user facts and agent self-narrative, and FSRS-based decay. On belief update, contradiction invalidation cuts stale-answer leakage from 80.0% to **34.0**%, at a moderate recall cost (update accuracy 92.0 → 76.0%) that exposes a precision/recall knob; we further show that the shipped lexical contradiction detector misses attribute-replacement updates and that a semantic detector recovers them (leakage 50.0 → **34.0**%). Namespace isolation reduces self-pollution from 96.8% to **1.6**% with no recall loss. Decay suppresses stale-trivia leakage from 100.0% to **16.4**% while retaining important facts. We release MnemoBench and all harness code.

1 Introduction

Memory systems for LLM agents (e.g. MemGPT/Letta [4], mem0 [1], Zep/Graphiti [5, 9]) are commonly assessed on multi-session question answering such as LoCoMo [2] and LongMemEval [7], which primarily stress *retrieval*. Yet the failures that erode trust in a long-lived assistant are different:

- **Belief staleness.** A fact changes (job, city, diet, relationship) and the system keeps surfacing the old value as if current — a failure mode recent work on knowledge updating in memory systems highlights as both common and hard [6].
- **Self-pollution.** In companion/agent settings the model emits first-person or speculative content; if mis-attributed it enters the user profile and is later returned as the user's own authoritative fact.
- **No forgetting.** Append-only stores accumulate one-off trivia that crowds out salient facts and lowers retrieval precision.

*Working draft. All reported numbers are from the frozen n=50 benchmark runs in the released harness.

We make two contributions. (i) **MnemoBench**, a benchmark isolating these three behaviours with generator-fixed ground truth (§4). (ii) An evaluation of the **Nemos** memory kernel (§3) showing each mechanism is responsible for a measurable, ablatable improvement (§6), including a before/after on a real defect we found and fixed in the contradiction detector.

2 Related Work

Memory architectures. MemGPT/Letta [4], mem0 [1], Zep/Graphiti's temporal knowledge graph [5, 9]. Contrast: layered separation, explicit invalidation semantics, source-authority gating.

Temporal/ bitemporal modelling. Graphiti edge validity intervals [9] vs. Nemos event-sourced invalidation with materialised flat fields.

Forgetting. FSRS / spaced-repetition retrievability [8, 3] applied to machine memory rather than human review scheduling.

Benchmarks. LoCoMo [2] and LongMemEval [7] are recall-centric. Closest to our thesis is the concurrent work of Uddin et al. [6], whose *Memora* benchmark and Forgetting-Aware Memory Accuracy (FAMA) metric explicitly penalise reliance on obsolete or invalidated memory — corroborating the recall-to-maintenance shift. MnemoBench is complementary: rather than a single aggregate score, it isolates each maintenance mechanism (invalidation, isolation, decay) through single-mechanism ablations, and additionally surfaces a before/after on a real detector fix.

3 The Nemos Memory Kernel

Nemos is an embeddable memory kernel. We describe only the mechanisms under test.

3.1 Layered memory and source authority

Five layers (episodic, semantic, personal_semantic, procedural, archival). Every record carries `source.authoritative` (user-stated vs. model-derived); derived content is barred from authoritative personal facts (invariant I4).

3.2 Bitemporal contradiction invalidation

Records carry world-axis (`valid_at`/`invalid_at`) and system-axis (`created_at`/`expired_at`) timestamps and a `belief_state` $\in$ {active, invalidated, superseded, corrected}. During offline reflection, a new fact that contradicts an existing `personal_semantic` anchor marks the old one `invalidated`; default retrieval returns only `active` records.

A real defect: lexical vs. semantic detection. The shipped detector prefilters contradiction candidates by character-bigram Jaccard similarity. This reliably catches strong negations ("left Google") but *misses attribute replacement* where the new value is lexically dissimilar (vegetarian $\rightarrow$ pescatarian). We replace the prefilter with embedding-cosine candidate retrieval and add a same-attribute, mutually-exclusive reconciliation pass over existing personal facts. §6 quantifies the before/after.

3.3 Namespace isolation (anti-self-pollution)

User facts live under `forUser(uid)`; agent self-narrative under `forUser(persona:pid)` — a physically separate keyspace. Retrieval of user facts cannot surface persona content.

3.4 FSRS decay

Each record has stability S, difficulty D, retrievability $R = \exp(-\Delta t/S)$. Access reinforces S; records with R below a cold threshold, zero recent access, after a dormancy window are marked cold and hidden from default search (archival is never cooled).

4 MnemoBench

4.1 Design principles

Ground truth is *fixed by the generator*: each item's fact script (which attribute, which values, in what order; which persona statements are traps) is decided first, then rendered into natural-language sessions. Scoring isolates *memory quality* from *generation quality*: an LLM judge inspects only the *set of retrieved records* and decides whether the expected fact is present and whether a forbidden (stale/leaked) fact is presented as current. The judge never writes the final answer.

4.2 Task families

BUC — Belief Update & Contradiction. A single-valued user attribute changes 1–3 times across sessions; the probe asks for the current value. *Update Accuracy* (expected present) and *Stale Leakage Rate* (a prior value presented as current; lower better).

ASP — Anti-Self-Pollution. User statements interleaved with persona self-statements, some crafted as plausible traps. Probes query user facts. *Pollution Rate* (persona fiction returned as authoritative user fact; lower better) and *User-Fact Recall*.

FOR — Forgetting & Salience. Durable important facts mixed with one-off trivia at older timestamps; salient facts are revisited during the retention interval. *Trivia Leakage* (cold trivia still surfacing; lower better) and *Important-Fact Retention* (guardrail; must stay high).

4.3 Item schema and scoring

Each item is a JSON record with an ordered list of `sessions` (`{speaker, text, contentDate?}`) and one or more `probes` (`{query, expected[], forbidden[]}`). For BUC the labels are derived deterministically from the fact-script (the last value is the only `expected` current value; all earlier values are `forbidden`), so the labels cannot drift from the rendered narrative. The judge receives the query, the `expected` and `forbidden` lists, and the retrieved record set, and returns two booleans — `contains_expected` and `contains_forbidden` — where a record that explicitly marks a forbidden value as past does *not* count as a leak.

Table 1: Belief update (BUC). Lower SLR is better; UA is the recall guardrail.

System	Update Acc. (%)	Stale Leakage (%)
mem0 (external)[1]	96.0	82.0
No invalidation (floor)	92.0	80.0
Lexical detector (shipped)	84.0	50.0
Semantic detector (this work)	76.0	**34.0**

Table 2: Anti-self-pollution (ASP) and forgetting (FOR).

ASP — Pollution Rate / User-Fact Recall (%)		
Shared store (baseline)	96.8	96.0
Namespace isolation	**1.6**	97.2
FOR — Trivia Leakage / Important Retention (%)		
Decay off	100.0	98.6
Decay on	**16.4**	97.9

5 Experimental Setup

All systems use OpenAI `gpt-4o` for extraction/reflection and `text-embedding-3-small` for embeddings; the judge is `gpt-4o`. We report $n = 50$ items per task. Each task is evaluated by ablating exactly the mechanism under test, so any difference is attributable to that mechanism:

- BUC: `semantic` (full) vs. `lexical` (shipped) vs. invalidation-off.
- ASP: namespace-isolated vs. shared store.
- FOR: decay-on vs. decay-off. To isolate decay we disable reflection in both arms and use cold threshold 0.3, dormancy 0, with a 120-day retention interval; important facts are accessed during the interval (modelling that salient facts are revisited).

As one external reference point we run mem0 (v2.0.8) on the BUC task with the same models and the identical judge. As a standard-benchmark cross-anchor we additionally evaluate a 30-question slice of the LongMemEval [7] *knowledge-update* subset (oracle setting), following its generate-then-judge QA-accuracy methodology, ablating invalidation (semantic vs. off).

6 Results

Belief update (BUC). Stale leakage falls monotonically across the ablation ladder: 80.0% with no invalidation, 50.0% with the shipped lexical detector, and **34.0**% with the semantic detector — the contradiction mechanism more than halves stale answers, and the lexical→semantic change alone removes a further third of the leakage by catching attribute-replacement updates the lexical prefilter misses. This gain is not free: update accuracy drops from 92.0% (floor) to 76.0% (semantic), i.e. aggressive invalidation occasionally retires a still-current fact. BUC thus exposes a precision/recall knob absent from the other two tasks; we regard the operating point as tunable (candidate-similarity

[1] mem0 v2.0.8, same gpt-4o + text-embedding-3-small, scored by the identical judge.

threshold, reconciliation conservativeness) rather than fixed. The external system, mem0, retains contradicted facts as active alongside their replacements (it appends without invalidating the old), placing its leakage near the no-invalidation floor.

External cross-anchor (LongMemEval). The same effect appears on a standard benchmark: on a 30-question slice of the LongMemEval knowledge-update subset (QA accuracy, generate-then-judge), enabling semantic invalidation raises accuracy from 33.3% to **43.3**% — a +10-point gain attributable solely to the invalidation ablation. The absolute level is modest (Nemos is used as a from-scratch memory over the oracle evidence sessions, without LongMemEval-specific tuning), so we read the *delta*, not the level, as the result: belief maintenance helps even on recall-style external data.

Anti-self-pollution (ASP). This is the largest and cleanest effect. Across 250 user-fact probes, namespace isolation reduces the pollution rate from 96.8% (a single shared store, as in append-only designs) to **1.6**% — a $\sim 60\times$ reduction — while user-fact recall is statistically unchanged (96.0% vs. 97.2%). Physically separating the agent's self-narrative keyspace from the user's facts essentially eliminates the failure mode at no cost to recall.

Forgetting (FOR). Over 146 salience probes, decay cuts trivia leakage from 100.0% to **16.4**% while important-fact retention is unchanged (98.6% vs. 97.9%): unreferenced trivia is cooled out of default retrieval without dropping the revisited important facts.

7 Discussion and Limitations

LLM-judge dependence. Both rendering and scoring use an LLM; we mitigate with a set-membership judge (no answer generation), low temperature, and a human-audited sample.

Self-generated data. Items are synthetic; we report only relative differences under identical data and add a standard-benchmark slice (LongMemEval) as a cross anchor.

FOR isolation choices. Disabling reflection and using aggressive decay parameters isolate decay cleanly but do not reflect the full-system default; we report them explicitly.

Residual leakage. The semantic detector does not reach zero stale leakage on generated hard cases (lexically and semantically close value pairs); we analyse failures in §6.

8 Conclusion

Long-lived memory should be judged on maintenance, not just recall. MnemoBench operationalises this, and each Nemos maintenance mechanism yields a measurable, ablatable gain.

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